

Constraint & Incentive Architecture Review

Challenge Point — Framework (Current Working Version) · 2026-06-17

The deliverable is a **framework** that clarifies how constraints and incentives shape the affordance landscape — **not** an expansion of the libraries. It organizes the existing 19-constraint inventory (Test Library V0) onto the shaping dimensions and reads directly off fields the rows already carry: `constraintRole`, `visibilityEffect`, `targetAffordancePrimary`, `includesIncentiveLayer`, `incentiveMechanism`. Incentives are not a separate library — they are the 13 constraints whose incentive layer is on.

The inventory, by role

The engine groups constraints into three roles and enforces a **role-mix rule**: every activity needs at least one foundation and one shaping constraint; consequence is optional. That rule is the spine of this framework — each role owns a different shaping dimension.

Role	#	Constraints	Owns
Foundation (structure)	6	Central Density, Zone Structure, Neutral Player, Goalkeeper Included, Small Area, Transition Trigger	Challenge + the base opportunity / information landscape
Shaping (hybrid)	4	Wide Zone Advantage, Counter-Press Window, Pass Combination Gate, Support Lane Requirement	Opportunity availability + information (sharpen a specific affordance; usually a light incentive)
Consequence	9	Progression Bonus, Wide Utilization Bonus, Switch of Play Bonus, Interception Reward, Turnover Reward, Delay Reward, Final Third Value, Transition Bonus, Recovery Window	Decision priorities (value)

Part 1 — Constraint Framework

The roadmap poses four questions. Each maps to a field already in the data.

Which constraints alter challenge?

The **foundation** constraints — they set the space / time / numbers / pressure baseline, i.e. the Environmental Fit dimensions:

Environmental Fit dimension	Expressed by
Space	Small Area Condition, Zone Structure Condition
Time	Small Area Condition, Transition Trigger

Environmental Fit dimension	Expressed by
Pressure	Central Density Condition, Small Area Condition
Numerical conditions	Neutral Player Condition
Information	see visibility, below
Transition volatility	Transition Trigger, Counter-Press Window

Which constraints alter information?

The **visibilityEffect** field encodes this directly: **indirect_increase** = structural reference points that make the picture readable (zones, central density, neutral overload, compactness); **increase** = raise the salience of a specific opportunity (Goalkeeper Included, and every incentive-bearing constraint); **neutral** = connects phases without adding a cue (Transition Trigger).

Which constraints alter opportunity availability?

The **targetAffordancePrimary** field names the affordance each constraint makes more reachable, and the foundation/shaping constraints open or close the space those affordances live in (Neutral Player opens overload; Small Area / Central Density compress it; Support Lane opens a penetration lane).

Which constraints alter decision priorities?

The **incentive layer** — constraints with an incentive on. By changing the value of options they shift what players weigh. This is the bridge to Part 2.

Takeaway: role predicts dimension. Foundation gives challenge + base opportunity/information; shaping sharpens a specific opportunity + its information; consequence tilts decision priorities. The engine's role-mix rule ("always a foundation + a shaping, consequence optional") is exactly: set the challenge and opportunity baseline first, then optionally tilt value. The framework validates that rule rather than changing it.

Part 2 — Incentive Framework

Incentives = the 13 constraints with an incentive layer. The **incentiveMechanism** field gives five mechanism types; each influences play differently.

Mechanism	Constraints	Primary influence
scoring_bonus (under-used action)	Switch of Play, Wide Utilization, Progression, Pass Combination Gate, Support Lane	Exploration — discover/try the rewarded option; attention to that cue
value_multiplier (target area)	Final Third Value	Exploitation — cash in where value already concentrates

Mechanism	Constraints	Primary influence
defensive_reward	Interception, Turnover, Delay	Tactical priority shift toward defending (regain / delay)
time_window_reward	Transition Bonus, Recovery Window, Counter-Press Window	Risk / tempo — commit-fast-vs-secure tension
positional_or_scoring_advantage	Wide Zone Advantage	Hybrid — opens an opportunity and rewards using it

Mapping to the five influence dimensions:

- **Exploration** — scoring_bonus on options players otherwise neglect (switch, width, combinations).
- **Exploitation** — value_multiplier on areas that already matter.
- **Attention** — every incentive's visibility lift: the reward is an attention magnet for its target affordance.
- **Risk** — time-window and contested rewards (Counter-Press Window: escape = bonus, fail = opponent advantage).
- **Tactical priorities** — the targetAffordancePrimary the incentive elevates in the team's priority order.

The ecological guardrail (the invariant)

Incentives raise the **value, attention, and visibility** of an opportunity. They must **not script the behavior**. Every row's contextualAudit enforces “invite, not force,” backed by the anti-scripting validators — the same principle as the recovery work: a solution becomes worth considering, never the only solution.

Open input — GF11 attacking-exploitation visibility

From Round 7.4: in the Recover & Reorganize environment the **attacking** race to exploit the disorder feels “more implied than explicit.” **Framework slot:** time_window_reward / Risk. The candidate incentive already exists — **Transition Bonus** is an attacker-side time-window reward — so this is a routing question, not an expansion. **Tested and deferred:** adding it makes it eligible but the scorer keeps it out (defensive consequences win on lens-alignment, and the top-2-per-bucket cap drops a 3rd-ranked cross-polarity incentive). Expressing a genuinely two-sided incentive contest needs an incentive-layer change — the first concrete problem for this review.

How to use this framework

- 1 **Start with challenge** — pick the foundation constraint(s) that set the right space/time/numbers for the goal and calibration.

- 2 **Shape the opportunity + information** — add a shaping constraint that makes the target affordance reachable and legible.
- 3 **Tilt decision priorities only if needed** — add a consequence/incentive to raise the value of the target affordance; pick the mechanism by the influence you want (explore vs. exploit vs. risk vs. defensive priority).
- 4 **Hold the guardrail** — the incentive must keep the problem a problem (multiple solutions), never collapse it into one prescribed action.

This is a lens for reasoning about the existing inventory; expansion should be driven only by gaps it reveals (e.g. the attacker-side two-sided-contest problem above), not by adding rows for their own sake.